Since x_1, x_2, x_3, and x_4 are positive integers, let us choose $t = 4$.

Then, we get $x_1 = 1, x_2 = 7, x_3 = 5$ and $x_4 = 4$.

So, the balanced equation is $C_5H_8 + 7O_2 \rightarrow 5CO_2 + 4H_2O$.

Example 1.40

If the system of equations $px + by + cz = 0,\ ax + qy + cz = 0,\ ax + by + rz = 0$ has a non-trivial solution and $p \neq a, q \neq b, r \neq c,$ prove that $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = 2.$

Solution

Assume that the system $px + by + cz = 0, ax + qy + cz = 0, ax + by + rz = 0$ has a non-trivial solution.

So, we have $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0.$ Applying $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$ in the above equation, we get

$$\begin{vmatrix} p & b & c \\ a-p & q-b & 0 \\ a-p & 0 & r-c \end{vmatrix} = 0. \text{ That is, } \begin{vmatrix} p & b & c \\ -(p-a) & q-b & 0 \\ -(p-a) & 0 & r-c \end{vmatrix} = 0.$$

Since $p \neq a, q \neq b, r \neq c,$ we get $(p-a)(q-b)(r-c)\begin{vmatrix} \frac{p}{p-a} & \frac{b}{q-b} & \frac{c}{r-c} \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0.$

So, we have $\begin{vmatrix} \frac{p}{p-a} & \frac{b}{q-b} & \frac{c}{r-c} \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0.$

Expanding the determinant, we get $\frac{p}{p-a} + \frac{b}{q-b} + \frac{c}{r-c} = 0.$

That is, $\frac{p}{p-a} + \frac{q-(q-b)}{q-b} + \frac{r-(r-c)}{r-c} = 0 \quad \Rightarrow \quad \frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = 2.$

EXERCISE 1.7

1. Solve the following system of homogenous equations.
   (i) $3x + 2y + 7z = 0,\quad 4x - 3y - 2z = 0,\quad 5x + 9y + 23z = 0$
   (ii) $2x + 3y - z = 0,\quad x - y - 2z = 0,\quad 3x + y + 3z = 0$
2. Determine the values of λ for which the following system of equations $x + y + 3z = 0,\ 4x + 3y + \lambda z = 0,\ 2x + y + 2z = 0$ has
   (i) a unique solution (ii) a non-trivial solution.
3. By using Gaussian elimination method, balance the chemical reaction equation:
   $C_2H_6 + O_2 \rightarrow H_2O + CO_2$

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